

Pharmacokinetic Simulation of Optimal Lopinavir and Ritonavir Dose Combination for COVID-19: Boosting Lopinavir With Ritonavir

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Abstract

Background/Aim: Lopinavir (LPV) combined with ritonavir (LPV/r) was initially developed to treat human immunodeficiency virus (HIV) infection and was subsequently repurposed to treat coronavirus disease 2019 (COVID-19) during the COVID-19 pandemic. As the efficacy of LPV/r in COVID-19 treatment has not been confirmed in clinical trials, LPV/r is not included in the Japanese COVID-19 treatment guidelines. Furthermore, previous clinical studies have not demonstrated the benefit of LPV/r against COVID-19 when used at the same dose as that used to treat HIV infection. Therefore, the aim of this study was to determine the optimal LPV/r dose combination for COVID-19 treatment.

Patients and Methods: Based on data from healthy volunteers and patients with HIV infection, maximum-effect models were used to estimate the relationship between LPV clearance and ritonavir plasma concentration. Pharmacokinetic simulations were performed using a range of assumptions based on previously reported modeling equations.

Results: The standard LPV/r dose combination of 400 mg/100 mg twice daily did not yield optimal blood concentrations. Based on the pharmacokinetic booster effect of ritonavir, the estimated optimal dose combination was 400 mg LPV boosted with 1,200 mg ritonavir.

Conclusion: These findings provide a basis to quantify the booster effect of ritonavir on LPV in COVID-19 treatment and calculate the optimal LPV and ritonavir dose combination.

Keywords: Pharmacokinetics, COVID-19, pharmacokinetic boosting, drug repurposing, simulation.

Introduction

The combination of lopinavir and ritonavir (LPV/r) was initially used for the treatment of human immunodeficiency

virus (HIV) infection and was repositioned for coronavirus disease 2019 (COVID-19) treatment during the pandemic (1, 2). However, the results of an open-label randomized controlled trial indicated that LPV/r is not a suitable



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treatment option for COVID-19 (1, 2); therefore, it is not included in the Japanese COVID-19 treatment guidelines (5th Edition) (3).

A previous study showed that lopinavir and ritonavir interacted well with the residues at the active site of SARS-CoV-2 protease and that this interaction appeared to play an important role in the binding of the drugs to the virus and their effect on the virus (4). Although ritonavir is also a protease inhibitor, it is used at a low dose in combination with LPV to inhibit the cytochrome P450 3A4 (CYP3A4) isoenzyme, which is responsible for the metabolism of LPV, leading to markedly increased plasma concentration of LPV (5, 6). In the open-label trial mentioned above, LPV/r was administered at a prescribed dose of 400 mg/100 mg twice daily, which is currently approved for the treatment of HIV infection but does not yield the blood concentrations required for ritonavir to have a pharmacokinetic (PK) booster effect (7). Currently, it is unclear whether LPV/r is effective against COVID-19; however, it is possible that the combination is not effective because sufficient blood levels are not maintained. Therefore, we focused on the PK parameters of LPV/r to test the hypothesis that the PK booster effect varies depending on the disease and then determined the optimal LPV/r combination dose for the treatment of COVID-19 using previously reported modeling equations (8).

Ritonavir at plasma concentrations of 0.06 and 0.36 mg/l has been associated with the half-maximal inhibitory concentration (IC_{50}) of LPV clearance at the standard 400 mg LPV/100 mg ritonavir twice-daily dosage in healthy volunteers and patients with HIV infection, respectively (9-11). Although information on the oral clearance (CL/F) of LPV boosted by ritonavir (the concentration at which this is achieved is considered the IC_{50}) is scant, based on a maximum-effect model in patients with COVID-19, we used a physiologically based pharmacokinetic (PBPK) model to predict the clinical effectiveness of the coadministration of lopinavir and ritonavir in COVID-19 treatment (12). However, Alvarez *et al.* (5) reported that the model for patients with COVID-19 used the IC_{50} value of healthy volunteers.

To our knowledge, no previous studies have reported a method to determine the IC_{50} of LPV/r for COVID-19 treatment. We developed a novel method to address this information gap. In this study, we used models developed by Thakur *et al.* (12) and Alvarez *et al.* (5) to predict the optimal dose combination of LPV and ritonavir in patients with COVID-19 by quantitatively assessing the PK booster effect of ritonavir on LPV, taking both therapeutic efficacy and safety into consideration.

Patients and Methods

Modeling process. Based on data from healthy volunteers and patients with HIV infection using LPV/r, maximum-effect models were used to estimate the relationship between LPV clearance and ritonavir plasma concentration.

The CL/F of LPV was estimated for healthy volunteers, patients with HIV infection, and patients with COVID-19 as follows (5, 9-11):

$$CL/F \text{ LPV} = CL_0 \times \text{inhibition},$$

$$\text{Inhibition} = 1 - [(I_{\max} \times RTC) / (IC_{50} + RTC)];$$

where, CL/F LPV refers to LPV clearance and CL_0 is 21.6, 11, 15.4, or 4.88 L/h, based on typical CL_0 values estimated using a PBPK model in studies involving volunteers and patients with HIV infection or those with COVID-19 (5, 9-11). $I_{\max}$ is the maximum inhibitory effect of ritonavir on CL/F, RTC is the trough concentration of ritonavir (mg/ml), and IC_{50} is the ritonavir concentration producing half of the $I_{\max}$. The IC_{50} value is typically used to measure drug efficacy; however, in this study, it was used as a measure of the PK booster effect of ritonavir.

Examination of the therapeutic ranges of LPV and ritonavir for COVID-19. The therapeutic dose ranges of LPV and ritonavir for COVID-19 have not been defined. Therefore, in addition to the criteria for efficacy reported previously (5), we considered criteria related to safety. A study on the safe concentration of LPV showed that it exhibited an anti-SARS-CoV-2 effect in Vero E6 cells at a half-maximal cytotoxic concentration (CC_{50} , which reduces the cell viability to 50%) of 49.75 μM *in vitro*,

Table I. Published studies related to the pharmacokinetics of lopinavir administered in combination with ritonavir.

Reference	Participants	CL ₀ (L/h)	CL/F (L/h)	Inhibition	IC ₅₀ (mg/l)
Dickinson <i>et al</i> , 2011 (9)	Healthy volunteers	21.6	1.16	0.05	0.057
Moltó <i>et al</i> , 2008 (10)	Patients with HIV infection	11	2.91	0.27	0.36
Chaivichacharn <i>et al</i> , 2022 (11)	Patients with HIV infection	15.4	2.64	0.17	0.207
Alvarez <i>et al</i> , 2020 (5)	Patients with COVID-19	4.88	0.60	0.12	0.057
Thakur <i>et al</i> , 2020 (12)*	Patients with COVID-19		6.02		

CL/F: Lopinavir clearance; CL₀: typical clearance; IC₅₀: half-maximal inhibitory concentration; PBPK: physiologically based pharmacokinetic model. *PBPK model.

corresponding to a blood concentration of approximately 31.3 mg/l (13). We used a reference value of 31.3 mg/l as the presumed safe concentration of LPV. In this study, we defined the therapeutic reference range by setting a criterion between the previously reported effective concentration (16.7 mg/l) and the presumed safe concentration (31.3 mg/l).

Heatmap of the appropriate LPV/r dose. A heatmap for determining the appropriate LPV/r dose was generated based on the increase in the LPV/r dose required to ensure efficacy and safety using IC₅₀ values of 1.0 and 4.0 mg/l. In addition, the appropriate LPV/r regimen was determined using the percentage of LPV concentrations that achieved the therapeutic reference range (16.7-31.3 mg/l).

Software. The data were analyzed using Phoenix NLME version 8.0 (Certara, Princeton, NJ, USA), and R version 4.1.2 (R Foundation for Statistical Computing, Vienna, Austria) was used for preparing the figures.

Results

Estimated IC₅₀ value of LPV in patients with COVID-19 from model simulations. The IC₅₀ of LPV for patients with COVID-19 was estimated from three previously reported IC₅₀ values (0.057 mg/l for healthy volunteers and 0.36 and 0.207 mg/l for patients with HIV infection) and a CL/F of 6.02 L/h predicted using the PBPK model (Table

I) (9-11). The IC₅₀ of ritonavir was estimated by first determining the ritonavir trough concentration (RTC) using the following procedure:

1) A linear regression was performed with IC₅₀ as the explanatory variable and CL/F as the dependent variable, and a correlation was observed (https://github.com/Nakamarusan/LPV_R_thesis).

2) The model used RTC values of 0.1 to 1.5 mg/l because the RTC has been reported to range from approximately 0.1 to 1.5 mg/l (9-11). A maximum-effect (E_{max}) value of 1 mg/l was used because previous studies used a value of 1 mg/l (fixed) or 0.929 mg/l (estimated) (5, 9). A typical CL₀ value of 15 L/h, based on the average of three values from healthy volunteers and patients with HIV infection, was used. The healthy volunteer data were fitted to a curve with an RTC of 0.6 mg/l, and data of patients with HIV infection were fitted to a curve with RTCs of 1.0 and 1.4 mg/l, respectively.

3) The LPV CL/F value for COVID-19 treatment determined using the PBPK model was 6 and the IC₅₀ values calculated from the intersection of the CL/F value with each RTC curve were 0.28, 0.41, 0.55, 0.68, 0.81, and 0.94 mg/l (Figure 1A).

4) A graph was created with the calculated IC₅₀, typical CL/F, and RTC for healthy volunteers and patients with HIV infection set to 0 to 2 L/h, with RTC on the horizontal axis and CL/F on the vertical axis. The IC₅₀ was determined to be 0.36 and 1.0 mg/l when the PBPK CL/F was 6 L/h using RTC values of 0.4 to 1.4 mg/l, respectively (Figure 1B).

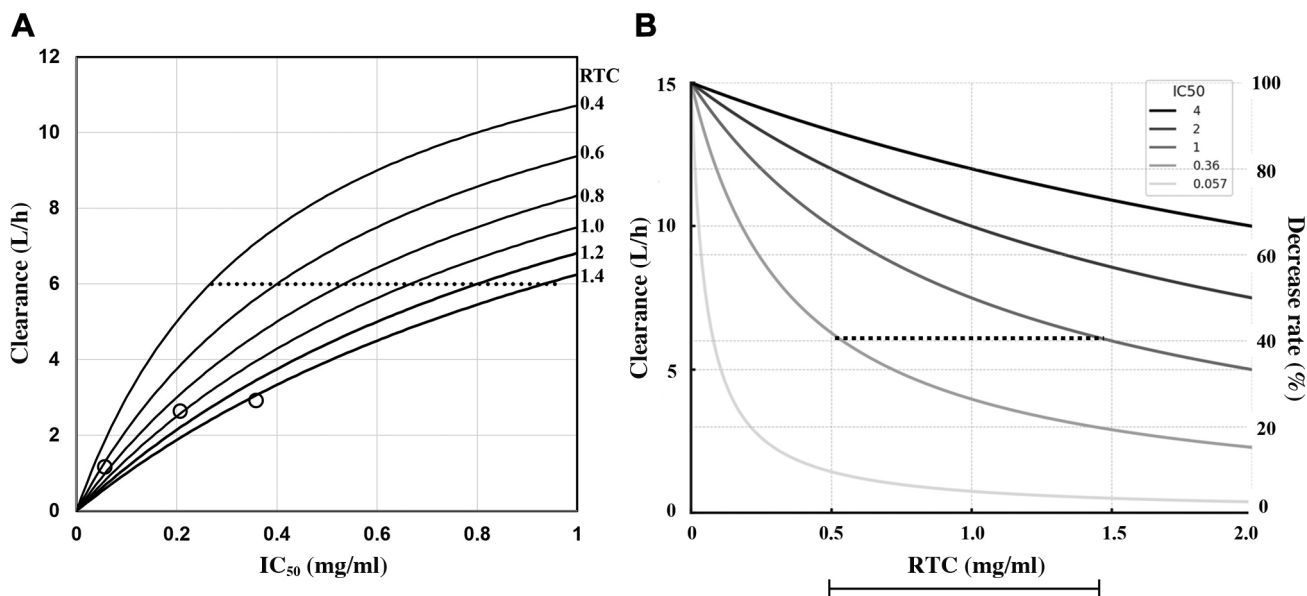


Figure 1. Estimation of the half-maximal inhibitory concentration (IC_{50}) of lopinavir and ritonavir in patients with COVID-19. (A) Correlation between IC_{50} and lopinavir clearance in healthy volunteers and patients with HIV infection. (B) Correlation between the trough concentration of ritonavir (RTC) and lopinavir clearance at each IC_{50} value. The dotted line indicates the oral clearance (CL/F) of lopinavir in patients with COVID-19 determined using the physiologically-based pharmacokinetic (PBPK) model.

Validation of the estimated IC_{50} value. Simulations were conducted for LPV/r, administered until LPV reached a steady state, using IC_{50} values of 0.057 and 0.36 mg/l for healthy volunteers and patients with HIV infection, respectively, and four different estimated IC_{50} values (1.0, 2.0, 3.0, and 4.0 mg/l). Subsequently, the actual measured values obtained from a previously published study, in which measurements were taken at approximately 120 h (3 patients) or 168 h (1 patient) in patients with COVID-19, were overlaid with the simulation results (Figure 2). The simulation with an IC_{50} of 4 mg/l aligned most closely with the actual measured values reported in the literature. Therefore, we inferred that an IC_{50} of 4.0 mg/l accurately represented the real-world situation. However, although this IC_{50} value resulted in an effective plasma concentration for treating HIV infection (0.07 mg/l), it did not attain the therapeutic range (16.7–31.3 mg/l) required for treating COVID-19.

The simulated results were visually inspected by overlaying clinical plasma concentration-time profiles (extracted using WebPlotDigitizer; San Francisco, CA, USA)

with model predictions. +, Δ , $\circ$, $\times$: clinical plasma concentration-time profiles.

Determination of the appropriate LPV/r dose in patients with COVID-19. When the dose of LPV was fixed at the prescribed dose (400 mg) using the IC_{50} value of 1.0 or 4.0 mg/l, increasing the dose of ritonavir caused the therapeutic reference range to exceed 40% at approximately 200 or $\geq 1,000$ mg, respectively. In contrast, when the dose of ritonavir was fixed at the prescribed dose (100 mg), increasing the dose of LPV caused the therapeutic reference range to exceed 30% at approximately 600 or $\geq 1,000$ mg, respectively (Figure 3). Therefore, the approved dose of LPV/r is insufficient for the treatment of COVID-19, and increasing the dose of the primary or PK booster drug may improve its efficacy. Our simulations showed that the PK profile fluctuation of the ritonavir dose range (mg/l) when the IC_{50} was 4 mg/l was smaller than that of the LPV dose range (mg/l) (Figure 3). An exploratory study of varying LPV/r dose combinations showed that the percentage of patients attaining the reference range was the highest when

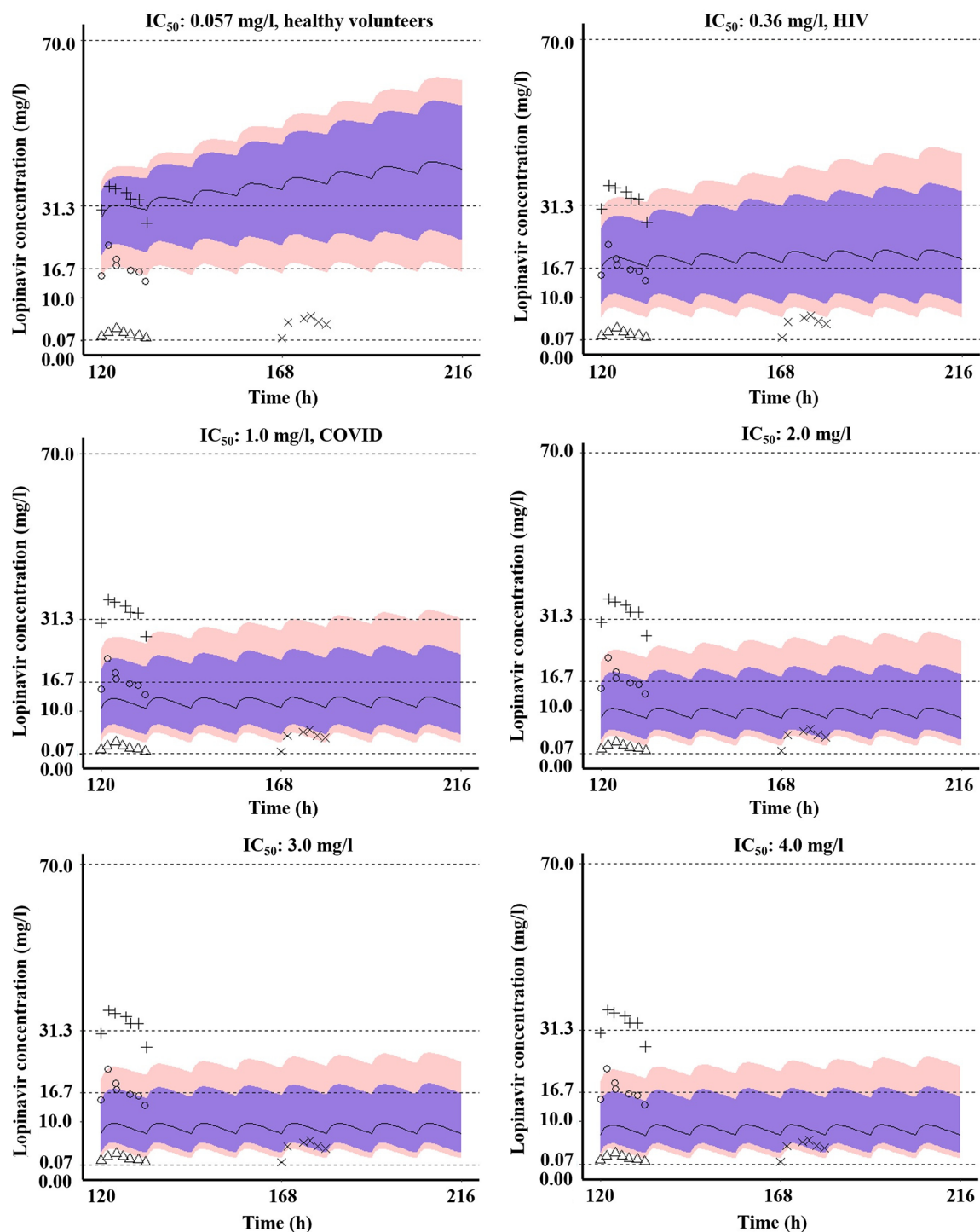


Figure 2. Simulated lopinavir (LPV) concentrations using the prescribed dose (LPV/ritonavir, 400 mg/100 mg twice daily) in patients with HIV infection or those with COVID-19. The solid line represents the median LPV concentration, and the shaded area represents the 90% prediction interval determined by 200 simulations. The upper and lower horizontal dashed lines represent the estimated safe concentration (31.3 mg/l) and effective concentration (16.7 mg/l) of LPV, respectively.

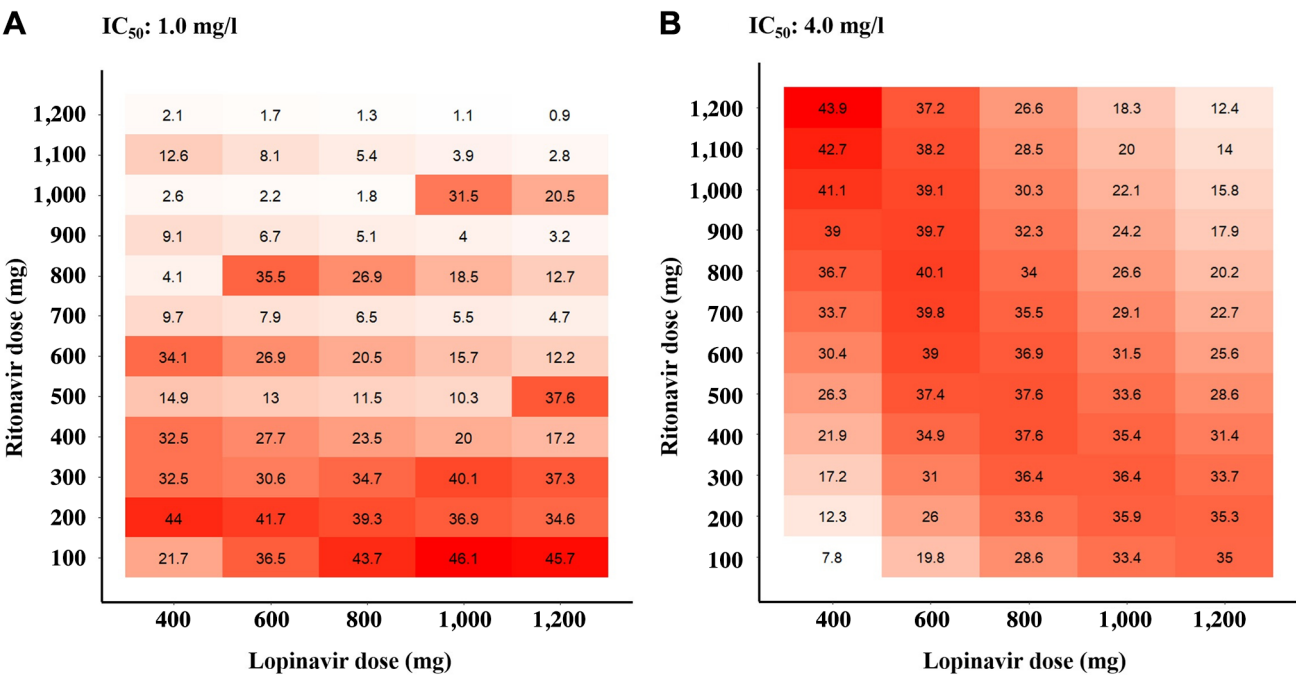


Figure 3. Heatmap showing the ratio of lopinavir (LPV) concentration covering the therapeutic reference range for various dose combinations. The reference lines indicate the LPV safe concentration (31.3 mg/l) and effective concentration (16.7 mg/l).

LPV/r was administered at 400 mg/1,200 mg. In contrast, when the IC_{50} was set at 1 mg/l, the percentage of patients attaining the reference range was the highest when the LPV/r dose was 1,000 mg/100 mg. These simulations suggest that increasing the dose of ritonavir is likely to improve the efficacy of LPV/r in patients with COVID-19 because a higher proportion of patients would achieve LPV concentrations in the reference range.

Discussion

In this study, the optimized dose of LPV in LPV/r for the treatment of COVID-19 was investigated by modeling the PK booster effect of ritonavir. To our knowledge, no previous study has quantitatively incorporated the PK booster effect into a model. In previous research (5), the concentration of combination drugs expected to have PK booster effects was fixed, and only the PK parameters of LPV were estimated. Most studies that examined the PK parameters of LPV have used a standard drug combination

at a ratio that has previously been evaluated in clinical trials. However, real-world data sometimes differ from the results of clinical trials, necessitating consideration of post-marketing data (1, 2). In such cases, it is useful to incorporate the PK booster effect into the model. Therefore, a similar strategy was used in this study. However, our simulations suggest that the PK booster effect does not occur in all scenarios. For example, in healthy volunteers, the IC_{50} was 0.057, indicating that the effect does not increase significantly even if the drug dose is increased.

First, the relationship between the IC_{50} of the drug and disease conditions was examined. A substantial percentage of healthy volunteers achieved effective concentrations at the prescribed dose, whereas in patients with COVID-19, the proportion of patients achieving effective concentrations at the prescribed dose was insufficient because the PK booster effect was different from that in healthy volunteers. In healthy volunteers administered the prescribed dose, the concentrations were well above the effective range; however, 74.4% of the

volunteers in the steady state exceeded the safety limit, raising concerns about potential adverse effects. The prescribed dose was insufficient to treat COVID-19; thus, we performed a dose-response sensitivity analysis. The strategy of increasing the ritonavir dose to 1,200 mg resulted in an optimal percentage of coverage within the effective range and plateaued with further increase.

The PK booster effect of ritonavir inhibits the activity of CYP3A4 and increases the concentration of LPV. As CYP3A activity in patients with COVID-19 is reduced by approximately 20% compared with that in healthy individuals (14), the LPV CL/F value should be lower and the blood concentrations of LPV should be higher in patients with COVID-19 than those in healthy volunteers. However, the CL/F value in patients with COVID-19 is higher than that in healthy volunteers. The exact mechanism underlying the irreversible inhibition of CYP3A by ritonavir is only partially understood, and various hypotheses and models for its exact nature have been proposed by different research groups (15). One of the potential mechanisms underlying the inhibitory effect of ritonavir is related to the inactivation of CYP3A by the formation of a ritonavir metabolic intermediate and reactive intermediate complex. It is considered that the amount of ritonavir metabolized by CYP3A4 decreases, resulting in a reduced inhibitory effect.

Pharmacometrics has been suggested as a robust approach to determine the application of drugs, including LPV/r, in the treatment of various diseases. This approach can be used to examine the optimal administration method, including the balance between the primary and PK booster drugs. For LPV/r, the PK booster effect is under quantitative investigation using a direct response $I_{\max}$ model (5), and the same model could be applied by just changing the model parameters such as IC_{50} and $I_{\max}$, depending on the relationship between the primary and PK booster drugs. Moreover, novel findings may be obtained using other mathematical models to interpret these data. For example, Paxlovid, a drug developed by Pfizer for COVID-19 treatment, incorporates ritonavir as a PK booster of nirmatrelvir (16). The quantitative examination of PK boosters, such as with Kaletra

(lopinavir/r), is expected to provide further information. In January 2022, Paxlovid (nirmatrelvir/r) was submitted to the Pharmaceuticals and Medical Devices Agency under the Special Approval for Emergency program (17).

In this study, we used an approach that combined bottom-up (PBPK) and top-down approaches, thereby enabling the development of more precise dosing recommendations. The simulation study used certain assumptions, and the results of simulations are sometimes distorted under conditions different from those in the real-world setting. Careful attention should be paid to environmental factors when interpreting the conclusions of this study, and the optimal LPV and ritonavir dosing combination should be considered under a range of conditions.

Conclusion

In conclusion, our pharmacokinetic simulation study identified a potentially optimal LPV/r dosing regimen for COVID-19 – 400 mg lopinavir combined with 1,200 mg ritonavir – which achieves higher predicted plasma concentrations of lopinavir compared to the standard 400/100 mg twice-daily regimen.

This study utilized both top-down and bottom-up approaches and attempted an integrated middle-out approach. Further research should expand and refine this middle-out methodology to improve simulation-based dosing strategies' accuracy and clinical applicability.

Conflicts of Interest

The Authors have no relevant financial or non-financial interests to disclose.

Authors' Contributions

YN, KS, and TM conceived and designed the study. YN and TM wrote the manuscript. NY, NI, KS, and TM analyzed the data. YM, FY and TM interpreted the results and contributed to the discussion. All Authors read and approved the final manuscript.

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